

No. of Functional Features Included in the Solution

Date	29 June 2026
Team ID	
Project Name	Rising Waters-ML Flood Prediction System
Maximum Marks	5 Marks

Functional Features Overview:

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	Home Page	Displays project introduction, objectives and navigation to prediction module.	Frontend (HTML/CSS/JS)	Completed	High
2	Flood Prediction Form	Accepts rainfall and weather parameters from the user for prediction.	Flask + HTML Form	Completed	High
3	Data Preprocessing	Performs feature scaling and preprocessing before sending data to the ML model.	Python (Scikit-learn)	Completed	High
4	Machine Learning Prediction	Uses the trained XGBoost model to classify flood risk.	XGBoost Model	Completed	High
5	Flood Chance Result Page	Displays warning message when flood risk is predicted.	Flask + HTML/CSS	Completed	Medium
6	No Flood Chance Result Page	Displays a safe message when no flood risk is predicted.	Flask + HTML/CSS	Completed	Medium
7	Model Persistence	Loads saved model (<code>floods.save</code>) and feature scaler (<code>transform.save</code>) without retraining.	Joblib	Completed	Medium
8	Responsive User Interface	Provides a clean and responsive web interface using HTML, CSS and JavaScript.	Frontend	Completed	Medium

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface	3	Home Page, Prediction Form, Prediction Result Pages
2	Machine Learning	2	Data Preprocessing, XGBoost Model Prediction
3	Backend Processing	1	Flask Routing , Input Validation and Model Integration
4	Model Management	1	Loading Saved Model and Feature Scaler
5	Frontend Design	1	HTML, CSS, JavaScript-based Interface